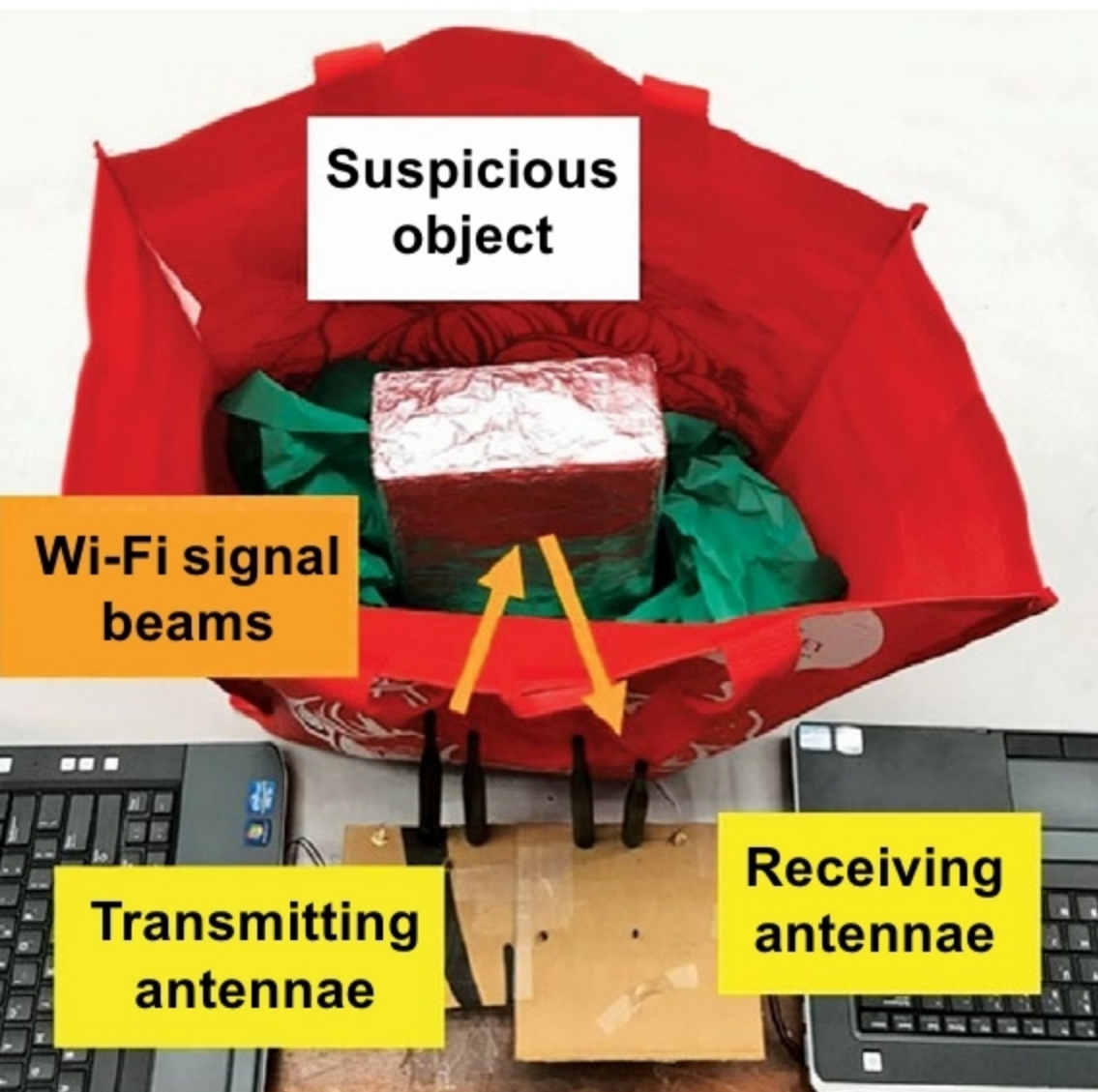


Ordinary Wi-Fi can detect weapons, bombs, explosive chemicals



According to a study led by Rutgers University–New Brunswick, common Wi-Fi can easily detect weapons, bombs and explosive chemicals in bags at museums, stadiums, theme parks, schools and other public venues. The suspicious object detection system is easy to set up, reduces security screening costs and avoids invading privacy such as when screeners open and inspect bags, backpacks and luggage. Whereas, traditional screening typically requires high staffing levels and costly specialised equipment.

Wi-Fi, or wireless, signals in most public places can penetrate bags to get the dimensions of dangerous metal objects and identify these, including weapons, aluminium cans, laptops and batteries for bombs. Wi-Fi can also be used to estimate the volume of liquids such as water, acid, alcohol and other chemicals for explosives, according to researchers.

This low-cost system requires a Wi-Fi device with two to three antennae, and can be integrated into existing Wi-Fi networks. The system analyses what happens when wireless signals penetrate and bounce off objects and materials.

Experiments with 15 types of objects and six types of bags demonstrated detection accuracy rates of 99 per cent for dangerous objects, 98 per cent for metal and 95 per cent for liquid. For typical backpacks, accuracy rate exceeded 95 per cent and dropped to about 90 per cent when objects inside bags were wrapped.

The discovery could have a great impact in protecting the public from dangerous objects (Credit: Rutgers University–New Brunswick)

Scientists develop world's most efficient semiconductor material for thermal management

Researchers at UCLA Samueli School of Engineering have developed a new semiconductor material—defect-free boron arsenide. It is more effective at drawing and dissipating waste heat than any other known semiconductor or metal material. This could potentially revolutionise thermal management designs for computer processors and other electronics, or for light-based devices like LEDs.

Managing heat in electronics has increasingly become one of the biggest challenges in optimising performance. This is because, as transistors shrink in size, more heat is generated within the same footprint. And, a lot of energy is used to keep those processors cool.

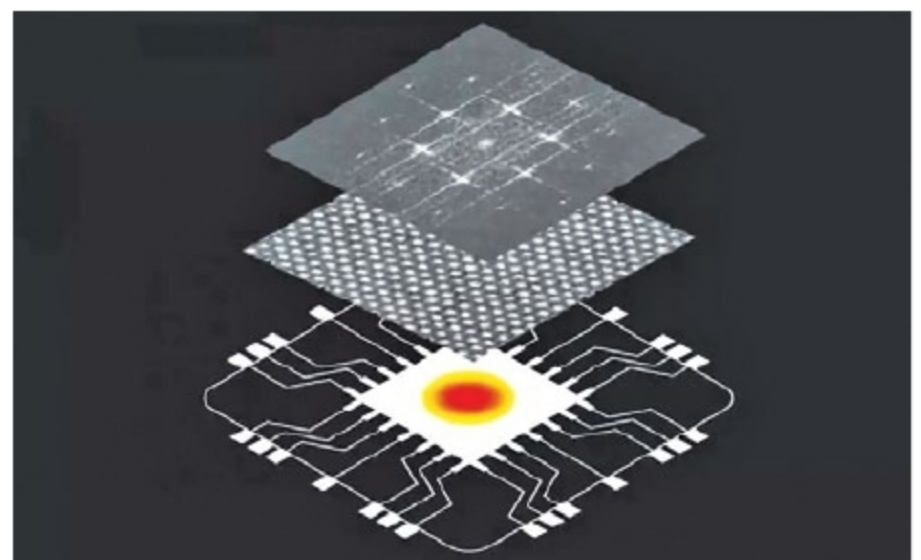


Illustration showing a schematic of a computer chip with a hotspot (bottom), an electron microscope image of defect-free boron arsenide (middle) and an image showing electron diffraction patterns in boron arsenide (Credit: Hu Research Lab/UCLA Samueli)